COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	10/21/26	**DATE SUBMITTED:**	10/21/26
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Activity: 6.1 **[ACTIVITY NAME]**

1. PROCEDURE OUTPUT

What procedures did you perform?

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
// Structure to store item details
struct Item {
    string name;
    int qty;
    double price;
};
int main() {
    // 1. Create file and write starter data
    ofstream outFile("inventory.txt");
    if (!outFile) return 1; // Stop if file fails to open
    outFile << "Notebook 50 25.5\nPencil 200 5\nEraser 80 7.25\n";
    outFile.close(); // Save and close
    // 2. Get new item from user and append it to the file
    Item newItem;
    cout << "Enter name qty price: ";
    if (cin >> newItem.name >> newItem.qty >> newItem.price) {
        ofstream appendFile("inventory.txt", ios::app); // Open in append mode
        if (!appendFile) return 1;
        appendFile << newItem.name << " " << newItem.qty << " " << newItem.price << "\n";
        appendFile.close();
    }
    // 3. Read file, print items, and calculate totals
    ifstream inFile("inventory.txt");
    if (!inFile) return 1;
    Item item, bestItem;
    double totalValue = 0, maxVal = -1;
    // Read item by item until end of file
    while (inFile >> item.name >> item.qty >> item.price) {
        cout << item.name << " " << item.qty << " " << item.price << "\n";

        double val = item.qty * item.price; // Item total value
        totalValue += val;                  // Add to grand total
        // Track the highest value item
        if (val > maxVal) {
            maxVal = val;
            bestItem = item;
        }
    }
    inFile.close();
```

```
Problems  Output  Debug Console  Ports  Terminal

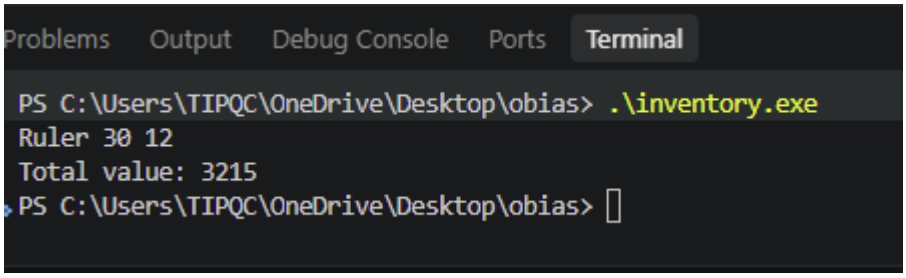
PS C:\Users\TIPQC\OneDrive\Desktop\obias> .\inventory.exe
Ruler 30 12
Total value: 3215
```

```cpp
    }
    inFile.close();
    // 4. Print final summary
    cout << "Total value: " << totalValue << "\n";
    if (maxVal >= 0) {
        cout << "Most valuable item: " << bestItem.name << " (Value: " << maxVal << ")\n";
    }
    return 0;
}
```

The program uses st

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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